

PROGRAM 6

Write a Python program to implement k-NN classification using the Iris dataset. Perform the following steps:

1. Load the dataset using Pandas.
2. Separate the features and target variable.
3. Split the dataset into 80% training and 20% testing data.
4. Apply StandardScaler for feature scaling.
5. Train a k-NN classifier with $k = 5$.
6. Predict the classes of the test data.
7. Display the confusion matrix, classification report, and accuracy.

PROGRAM 7

Write a Python program to implement k-NN classification using the Cancer dataset.

PROGRAM 8

Experiment: Wine Quality Analysis and k-NN Classification

Dataset: Wine_Quality.csv

Write a Python program to:

a) Perform Exploratory Data Analysis (EDA):

- Display the first five records.
- Check for missing values.
- Display descriptive statistics.
- Display the distribution of wine quality.

b) Visualize the data using:

- Scatter plot of alcohol vs. pH.
- Histogram of alcohol.
- Box plot of alcohol grouped by quality.

c) Implement k-NN Classification:

- Remove missing values and prepare features and target.
- Split the data into 80% training and 20% testing sets.
- Standardize the features using StandardScaler.
- Apply k-NN with different values of k and display the accuracy.

- Allow the user to enter k values and display the corresponding accuracy.